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Review

Linking Drift Prediction to Search Planning in Maritime Search and Rescue: A Review of Cross-Stage Evaluation

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Abstract

Maritime search and rescue requires probabilistic decision-making under uncertainty about a drifting target's location and constraints on search resources. Classical search theory provides a basis for allocating search effort through the probability of containment, probability of detection, and probability of success. Advances in numerical modeling and artificial intelligence have expanded the capabilities of drift prediction and search planning, but improvements in prediction accuracy do not necessarily translate into better search outcomes. This structured narrative review examines the development of these research streams and their coupling through probability representations, including probability-of-containment maps. An illustrative evidence map of ten selected cases distinguishes implemented coupling from evidence that isolates the downstream effects of prediction changes. The cases provide limited evidence tracing these effects through probability representation, planning, and operational outcomes under common evaluation conditions. The synthesis organizes evaluation into four stages: Prediction Gain, Information Gain, Decision Gain, and Operational Gain. Decision Gain denotes improved plan quality under a common evaluation criterion, rather than a change in the plan alone. The review outlines common evaluation conditions and a factorial comparison for separating prediction-side, planning-side, and combined effects. These recommendations support evaluation of whether artificial-intelligence-based prediction improvements lead to better search decisions, detection outcomes, and resource use.

Keywords: maritime search and rescue; search theory; probability of containment; probability of detection; probability of success; drift prediction; artificial intelligence; path planning; resource allocation; decision support

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1. Introduction

The central challenge in maritime search and rescue (MSAR) is to locate a target whose position is uncertain, using limited time and resources. A person in distress at sea continues to drift under the influence of currents, wind, waves, and tides. Thus, even when the last known position is available, the target's location at the time of the search is naturally represented by a spatial probability distribution rather than a single coordinate. Classical search theory established a basis for allocating limited search effort through the probability of containment (POC), the probability that the target is within a specified area; the probability of detection (POD), the conditional probability of detecting the target when that area is searched; and the probability of success (POS), which combines these two quantities [1–8]. The International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual and the US Coast Guard's Search and Rescue Optimal Planning System

(SAROPS) translate this probabilistic perspective into operational procedures and decision support [9–11].

Drift prediction has developed as a separate research stream. Early studies focused on deterministic trajectories derived from empirical leeway relationships and numerical ocean models. The introduction of Monte Carlo particles, data assimilation, and ensemble methods enabled the generation of location distributions that disperse over time [12–16]. As Lagrangian observations, including those from the Global Drifter Program, and environmental fields such as the fifth-generation European Centre for Medium-Range Weather Forecasts reanalysis (ERA5), the Hybrid Coordinate Ocean Model (HYCOM), and the Global Ocean Reanalysis and Simulations (GLORYS) became widely available [17–24], statistical models, machine learning (ML), deep learning (DL), hybrid approaches combining physics and artificial intelligence (AI), and probabilistic prediction were introduced to reduce prediction errors and computational costs [25–33].

Search planning has also advanced rapidly. Standard patterns, including parallel sweep, creeping line, expanding square, and sector search, remain operationally important. Meanwhile, probability-based search-area allocation, mathematical optimization, genetic algorithms (GAs), particle swarm optimization (PSO), reinforcement learning (RL), deep reinforcement learning (DRL), and cooperative planning for multiple unmanned aerial, surface, and underwater vehicles (UAVs, USVs, and AUVs) have expanded the scope of automated decision-making [34–50]. Recent work has incorporated practical operating conditions through GA-based search-area deployment with civilian vessel integration and probability-based planning subject to land constraints [51,52].

Drift prediction and search planning address different research questions and therefore appropriately use different evaluation metrics. Prediction studies typically assess trajectory accuracy using mean absolute error (MAE), root mean square error (RMSE), normalized cumulative Lagrangian separation (NCLS), and skill scores, with increasing attention to distributional quality [25,27,33,53]. Planning studies evaluate coverage, path length, computation time, reward, resource cost, or POS [37,38,45,46]. The central question is whether sufficient evidence connects improvements in drift prediction to subsequent probability representation, search planning, and operational outcomes. For example, a reduction in mean position error may leave search and rescue unit (SRU) deployment and routes unchanged if the actual target remains in a low-probability tail or the same search area is generated. Conversely, a modest improvement in position error may substantially change the plan if the ranking and probability mass of high-probability regions better reflect the target's actual location.

Existing reviews establish the background for the present synthesis: Breivik et al. cover developments in drift modeling and search theory [15]; Kim et al. organize surface-drifter prediction from numerical models to artificial intelligence [33]; and Kilic et al. review artificial-intelligence-based maritime search and rescue planning across environmental estimation, resource allocation, and execution [54]. The present review addresses a narrower comparison: what evidence is needed to attribute a downstream change in search performance to an upstream prediction change? It distinguishes probability representation, plan quality, and execution outcomes; maps the intervention scope of selected integrated cases; and specifies common evaluation conditions for separating prediction-side, planning-side, and combined effects. Its contribution is this MSAR-specific organization of cross-stage evidence, rather than a claim that the distinction between forecast accuracy and decision value is new.

Review Scope and Literature Selection

This study was designed as a structured narrative review of the links between drift prediction, probability representation, and search planning, rather than a systematic review or meta-analysis of effect sizes. The scope included research available by 14 September 2026. Topic-based searches used "maritime search and rescue", "drift prediction", "leeway", "probability of containment", "probability of detection", "probability of success", "search planning", "coverage path planning", "resource allocation", "reinforcement learning", "probabilistic forecast", and combinations of these terms, together with backward and forward citation tracing. The reviews by Breivik et al., Kim et al., and Kilic et al. served as the main seed publications [15,33,54], supplemented by foundational studies and recent work from 2024–2026.

The literature was organized into four categories: drift/leeway and probabilistic prediction studies directly applicable to MSAR; MSAR planning studies addressing search-area allocation, coverage or path planning, and resource allocation; integrated MSAR studies explicitly connecting prediction or probability representation to planning; and the adjacent enabling literature on probabilistic forecast verification, forecast value, prediction-to-decision learning, data assimilation, and weather/ocean AI. General nonmarine robotic search, image-based target detection, medical rescue, and disaster-response studies were excluded unless they directly informed the probability-based prediction–planning interface in MSAR. The adjacent literature was used to support evaluation theory and future technical directions, rather than as direct evidence of MSAR performance.

The final bibliography comprises 98 sources, including peer-reviewed articles, books and technical reports, operational guidance, and authoritative model and data documentation. The evidence synthesis maps ten purposively selected cases across Prediction (P), Information/Probability representation (I), Decision/Planning (D), and Operational outcome (O). Inclusion and validation are assessed separately: implementing a connection does not establish the effect of changing its input. Conclusions are restricted to the selected evidence and do not estimate validation frequency or establish research absence across the field. Complete platform-specific search strings and screening records are not provided, so full search reproducibility and exhaustive coverage are not claimed.

Each selected case links at least two adjacent stages and describes the information passed between them. Selection covers complementary interfaces: ensemble-to-area construction, operational decision support, Bayesian updating, evolutionary deployment, probability-guided planning, and land-constrained allocation. A complete predictor-intervention comparison is not required for inclusion. The cross-stage evidence map records the implemented stages, evaluation basis, and intervention evidence, with source locations where identified. This purposive coverage does not establish statistical representativeness. The assessment concerns evidence for attributing cross-stage effects, not the overall quality of each study.

Section 2 reviews drift prediction from physical models to AI and uncertainty-aware approaches, with emphasis on what has been evaluated. Section 3 examines search planning from classical search theory and SAROPS to optimization, RL/DRL, and multi-agent search, comparing the decisions and metrics associated with each method family. Section 4 examines probability representation and integrated frameworks to identify the extent of validation along the prediction-to-planning pipeline. Section 5 derives and addresses the research questions, and Section 6 discusses future directions, including end-to-end benchmarks and search-oriented probabilistic evaluation.

2. Development of Drift Prediction for Maritime Search and Rescue

This section examines the inputs, models, outputs, and evaluation practices of successive developments in drift prediction. Kim et al. organized this development into numerical modeling, data assimilation, statistical/probabilistic methods, ML/DL, hybrid approaches, and AI-based assimilation [33]. Here, the discussion centers on the uncertainty or probability information that each method can supply to downstream search planning and the metrics used to evaluate that information.

2.1. Classical Physics-Based and Leeway Drift Prediction

Early drift models predicted future positions by integrating motion driven by surface currents, wind, waves, and other environmental forcing. Surface Velocity Program drifters became a principal observational tool for estimating surface currents and validating numerical models [17–19], while expanding observation networks supported model calibration and statistical learning [20,21]. Following the Prestige oil spill, drifting buoys were used to track the spill, and numerical wind and current fields were combined with Lagrangian transport models to predict and refine dispersion trajectories [55,56]. Subsequent oil-spill models increasingly integrated atmospheric and oceanic forcing, in situ data assimilation, and uncertainty representation [57].

In MSAR, leeway is a key variable because a search object does not simply follow the current but also experiences wind-induced drift. Breivik and Allen presented an operational search and rescue (SAR) model for the Norwegian Sea and the North Sea in northwestern Europe that combined currents, wind, waves, and object-specific leeway [12]. Standardized leeway field methods subsequently enabled empirical coefficients and uncertainties for different object types to be incorporated into models [13]. Breivik et al. later developed BAKTRAK, a backtracking method that repeatedly applies the same forward drift model to infer an initial location from observations [14]. OpenDrift implemented Lagrangian particle modeling in a general framework, providing common readers and stochastic modeling tools for SAR, oil spills, and other drift applications [16].

These approaches offer interpretable representations of forcing and object physics. However, small errors in initial conditions, environmental-field resolution, and leeway coefficients accumulate over long forecasts. A single representative trajectory also provides an incomplete description of the location uncertainty required for MSAR [15,58]. These limitations motivated the broader use of data assimilation and ensemble/probabilistic modeling.

Before drift integration, multisource inputs require a defined interface: wind, current, wave, and observational products should be aligned to a common time reference, spatial coordinates, units, and relevant depth or height, with interpolation, missing-data handling, and product latency documented. Numerical forcing can then drive the particle model, while observations may correct its state through assimilation; these operations should be distinguished from learned feature fusion. OpenDrift's readers and the reviewed assimilation methods illustrate these different roles [16,59,60]. Learned fusion offers an additional option: Huang et al. combine multisource remote-sensing information for urban impervious-surface extraction [61], an adjacent methodological example rather than evidence of improved MSAR drift prediction. For cross-stage comparisons, changes in input sources or fusion should be treated as explicit interventions, and any resulting changes in forecast uncertainty should be carried through to the probability map.

2.2. Data Assimilation and Statistical and Probabilistic Drift Prediction

Long-range Lagrangian trajectory prediction is challenging because ocean dynamics are nonlinear and sensitive to initial conditions. Özgökmen et al. examined the relationship

between Lagrangian predictability and drifter density in an idealized ocean [62]. Castellari et al. showed that Gauss–Markov and Kalman-based corrections could reduce prediction errors for Adriatic Sea drifters [63]. Molcard et al. reconstructed Eulerian circulation fields from drifter observations [64]. Muscarella et al. and Sun and Penny examined improvements in Lagrangian predictability from assimilating drifter velocities or positions into operational ocean models or ensemble Kalman frameworks [59,60]. Holm et al. demonstrated the computational feasibility of directly representing uncertainty in drift forecasting through a massively parallel particle filter [65].

A major development was the expansion of model outputs from single trajectories to ensemble spread and probability density. Monte Carlo particles sample uncertainties in initial location, forcing, and leeway to generate possible future states, providing a natural basis for search-area generation. Environmental-field quality also became increasingly important. High-resolution reanalyses such as ERA5 and ocean fields from HYCOM and GLORYS are widely used in both Lagrangian integration and ML models [22–24]. More recent work has fused drifter observations with numerical outputs to improve estimates of turbulent dispersion [66].

Statistical and probabilistic approaches are well suited to MSAR because they describe both the expected location and its uncertainty. Nevertheless, evaluation during this period often emphasized mean distance error, RMSE, ensemble spread, or selected confidence regions, without separately assessing the actual target's search priority within the predicted distribution.

2.3. Introduction of Machine Learning and Early Data-Driven Prediction

From the late 2010s onward, the increasing availability of drifter observations, reanalysis data, and graphics processing unit (GPU) computing resources supported the application of ML to drift prediction. Using Korean coastal drifter data, Nam et al. compared learning and evolutionary methods, including support vector regression (SVR), radial basis function networks (RBFNs), long short-term memory (LSTM) networks, differential evolution (DE), PSO, and covariance matrix adaptation evolution strategy (CMA-ES), and showed that some models achieved lower MAE or higher NCLS-based skill scores than conventional numerical models [25]. Aksamit et al. combined physical modeling with data-driven correction by training an LSTM to learn residual errors in physics-based trajectories [26]. Grossi investigated graph- and convolutional neural network (CNN)-based approaches to surface-drifter trajectory prediction [67]. In the adjacent field of inland-water drift, Zeng et al. applied deep learning to buoy time series to improve short-term prediction [68].

ML was primarily used to learn nonlinear relationships between observations and environmental variables or to correct existing forecasts. GAs also supported optimization within prediction pipelines, including oil-spill particle matching and feature selection [69,70]. Other approaches estimated empirical parameters such as the wind drift factor using ML [71], or used lightweight wind–current weighted models to maintain computational efficiency and interpretability [72]. The studies reviewed in this category primarily reported improvements in position prediction accuracy using MAE, RMSE, and NCLS.

2.4. Deep Learning and Spatiotemporal Drift Prediction

Convolutional and recurrent networks, attention mechanisms, and hybrid architectures have been adopted to process large environmental fields and trajectory datasets jointly. Botvynko et al. proposed a fully convolutional DriftNet inspired by the Eulerian Fokker–Planck representation to learn sea-surface Lagrangian drift [27,28]. Ning et al. combined multiscale ocean drivers with time–space attention for Argo trajectory prediction [73]. Song et al. fused environmental variables using CNNs, bidirectional gated recurrent units

(BiGRUs), and attention [29], while Chen et al. introduced target-area differential attention and a direction–distance loss to improve marine drifter prediction in the Taiwan Strait [30].

Beyond direct trajectory regression, surrogate models have been developed to predict spatial fields and particle patches. Fajardo-Urbina et al. presented a deep-learning surrogate for coastal particle transport, aiming to accelerate inference relative to conventional Lagrangian particle models [74]. Bang et al. reconstructed surface velocity fields from drifter data using a physics-informed neural network [75]. Pan et al.'s space–air–sea–ground monitoring framework integrated multiple observation sources and forecasting systems, illustrating the potential of multisource data fusion in marine incidents [76]. Although Domala et al. did not directly study SAR, their wave-prediction work provides an adjacent example of how data-driven wind and wave forecasts could improve upstream forcing for drift prediction [77].

Evaluation at this stage began to include long-term trajectory consistency, directional error, and computation time alongside pointwise accuracy. However, many of the reviewed studies continued to compare models primarily through the distance between the predicted center or representative trajectory and the observations.

2.5. Hybrid and Uncertainty-Aware Prediction

Recent work increasingly combines physical consistency with uncertainty representation. Jenkins et al. trained a deep neural network (DNN) to predict the trajectory probability density function (PDF) at the next time step, bringing ensemble-like distributional outputs into a neural framework [31]. Brolly estimated ocean transport statistics using a probabilistic neural network [32]. Beiser studied local probabilistic drift forecasting by combining simplified ocean models with multilevel data assimilation [78]. Lin et al. analyzed the effects of ocean-current features on ML and dynamic tracking performance, highlighting regional dependence [79].

In broader ocean–atmosphere forecasting, AI has begun to enter data assimilation and the forecast cycle itself. Sun et al. proposed an observation-to-forecast ML system [80]; Adrian et al. analyzed the integration of a FourCastNet surrogate with conventional data assimilation [81]; and Allen et al. presented an end-to-end weather system that produces global forecasts directly from observations [82]. These approaches have not become established standards for MSAR drift prediction, but they indicate changes in the upstream infrastructure that could generate ocean forcing and trajectory uncertainty in real time.

A 2026 OpenDrift study in the southern coastal waters of Korea illustrates the growing emphasis on search-oriented evaluation. In addition to RMSE and NCLS, it used Location Prediction Conformance (LPC), indicating whether the observed location lies within the predicted area, and the Conformance–Effort Ratio (CER), which relates containment to search area [83]. This extends evaluation beyond representative-trajectory accuracy to the trade-off between target containment and the required search area.

2.6. Evolution of Drift Prediction Metrics

Drift prediction metrics have expanded from pointwise position errors to trajectory consistency, distributional quality, and operational containment. The NCLS measure of Liu and Weisberg normalizes cumulative trajectory separation by observed travel distance to assess trajectory reproduction over time [53]. It has been widely used alongside MAE and RMSE in ML/DL studies [25,27,33]. Normalized separation must be distinguished from a skill score derived from it. For example, the NCLS-based skill score used by Nam et al. improves as it approaches one; the direction of improvement must therefore be interpreted according to each study's definition [25]. Mean error alone cannot assess both calibration

and sharpness in a probabilistic forecast. General forecast-verification methods include rank histograms, calibration diagnostics, sharpness measures, proper scoring rules, and the energy score [84–86]. Highest density regions (HDRs) provide a classical way to summarize a distribution by preferentially including locations with higher density, which is useful for multimodal spatial distributions [87].

Table 1 summarizes these changes in evaluation emphasis. New metrics have generally complemented earlier metrics by detecting different failure modes.

Table 1. Development of drift prediction methods and associated evaluation metrics.

Research Stream	Main Outputs	Typical Metrics	Evaluation Focus
Physics-based deterministic prediction	Representative trajectory	Distance error, MAE, RMSE	Proximity of the representative prediction to observations
Data assimilation/ensembles	Ensemble trajectories and spread	RMSE, spread, confidence regions	Initial-state correction and uncertainty extent
Early ML/DL	Learned trajectory	MAE, RMSE, NCLS, skill score	Pointwise accuracy and trajectory reproduction
Spatiotemporal DL/hybrid methods	Trajectory, field, or surrogate output	NCLS, directional error, runtime	Long-term stability, direction, and computational efficiency
Probabilistic and operational approaches	PDF, particle cloud, search area	Likelihood, calibration, HDR, LPC, CER	Distributional reliability, containment, and required search area

3. Development of Maritime Search and Rescue Planning

This section examines the development of search planning independently of drift prediction. Given a common target distribution, the planning question is how much effort to allocate to each area and which routes the SRUs should follow. Classical probabilistic search theory defined objectives directly through POC, POD, and POS. Optimization and AI subsequently introduced additional proxy metrics, including coverage, reward, energy, and path length. The emphasis here is on the probability information used by each method family and the downstream decisions or outcomes it evaluates, complementing existing algorithm-oriented reviews [54].

3.1. Classical Search Theory and Operational Planning Systems

Koopman's search theory formalized the relationship between detection probability and search effort and the optimal spatial allocation of limited effort [1–4]. It was subsequently extended to optimal search frameworks incorporating moving targets, imperfect detection, stopping decisions, and Bayesian updates [5–8]. If POC denotes the probability that a target is within a given area and POD denotes the conditional probability of detecting it when that area is searched, the success probability for the area is given by Equation (1)

$$\mathrm{POS} = \mathrm{POC} \times \mathrm{POD}. \tag{1}$$

For multiple grid cells i, with target probabilities p_i and conditional detection probabilities d_i, the success probability of a plan can be written as Equation (2)

$$\mathrm{POS} = \sum_i p_i d_i. \tag{2}$$

This is a static, discretized notation used to explain the prediction–planning interface. Here, p_i is the target probability of a mutually exclusive cell, and d_i is the probability of detection under the specified plan, conditional on the target being in cell i. In dynamic

MSAR, both $p_i(t)$ and $d_i(t)$ can change with target motion, search timing, and updates following unsuccessful searches. Equation (2) therefore provides a simplified interface representation of the sequential Bayesian setting discussed below.

IAMSAR translates search theory into operational procedures through sweep width, track spacing, coverage factors, and standard search patterns [9]. SAROPS generates search-object probabilities using environmental forcing and Monte Carlo particles, then supports planning by incorporating prior search effort and unsuccessful searches through Bayesian updates [10,11]. POC, POD, POS, and search effort thus formed the operational performance language of these classical systems from the outset.

3.2. Resource Allocation Using Mathematical Optimization and Metaheuristics

Search-area allocation becomes a combinatorial optimization problem when multiple SRUs, limited search time, and different speeds and detection capabilities are involved. Otote et al. presented a maritime SAR decision algorithm using POC, POD, and POS [34]. Burciu analyzed how improved detection capability affected the success probability of SAR operations [88]. PSO-based time–space weighting was used to identify efficient allocations for helicopter search planning [35], followed by enhanced PSO and multi-objective approaches to resource allocation [40,41,46]. Other work incorporated accident black spots and historical risk information into allocation decisions [89].

GAs and evolutionary computation are suitable for search-plan design involving mixed continuous and discrete variables, including area location, dimensions, orientation, and coverage parameters. Hong and Kim optimized maritime SAR search-area deployment and civilian vessel collaboration using a GA, using POD-adjusted fitness to account for the detectability of drifting targets [51]. Jeong et al. further developed a time-dependent weighted-heatmap framework in which Monte Carlo drift particles were converted into probability-density representations for GA-based SRU deployment, followed by dynamic search-plan evaluation using cumulative POS [90]. Zhan et al. combined evolutionary search with learning for UAV path planning [47]; Wu et al. applied RL to maritime resource allocation [49]; and Zhang et al. used multi-objective optimization for multi-UAV coverage planning to address high-priority area coverage, workload balance, path safety, and energy [50]. These studies address multiple operational objectives within complex decision spaces.

3.3. Probability-Map-Based Coverage and Path Planning

Beyond allocating search areas, planners must determine the order in which an SRU visits cells or regions. Ren et al. generated POC maps from Monte Carlo particles, updated probabilities after searches using Bayes' rule, and planned routes that prioritized high-probability regions [36]. Ai et al. introduced RL into maritime SAR coverage path planning to incorporate environmental information and search priorities [37]. Wu et al. combined person-in-water drift prediction with a hierarchical probability map for autonomous coverage path planning [44]. Zhang et al.'s UAV framework also directly coupled drift/Monte Carlo probability information with coverage planning [48].

In these approaches, the probability map is a key input to the planner's objective or reward. Consequently, the same planning algorithm may concentrate search effort in substantially different locations depending on the sharpness and calibration of the input map.

3.4. Path Generation Using Reinforcement and Deep Reinforcement Learning

RL/DRL can directly handle complex states and dynamic environments and has therefore been rapidly adopted in MSAR path planning. POS-DQN incorporated POC, POD, and POS into deep Q-network (DQN)-based USV path planning, directly connecting

classical search metrics to a learning objective [45]. In multi-AUV search, multi-agent DRL was used to learn cooperative policies for a moving, unobserved target [42]. RL-based evolutionary UAV planning combined reward learning with population-based search to address limitations of standalone RL and metaheuristic approaches [47].

RL allows a policy to select actions sequentially in response to search results and environmental changes. However, the interpretation of performance depends strongly on reward design: a higher episodic reward does not necessarily imply a higher POS. Explicitly relating rewards to classical SAR metrics is therefore important.

3.5. Multi-Agent and Heterogeneous Autonomous Search

MSAR operations often involve cooperation among several aircraft, vessels, UAVs, USVs, or AUVs. Cho et al. addressed maritime SAR coverage by multiple UAVs using grid decomposition and optimization [38,39], while Cai et al. proposed coverage planning for multiple AUVs [43]. Multi-agent DRL provides one approach to moving-target search and decentralized coordination [42]. Recent resource-allocation studies have sought to account jointly for SRU capabilities, arrival times, workloads, and mission priorities [46,49].

In heterogeneous systems, POD depends on differences in sensor footprint, platform speed, endurance, altitude or depth, and communication constraints. Total coverage and path length alone are therefore insufficient for comparing different SRU combinations; capability-aware detection models and cumulative POS are also needed. Saito's 2026 study linked leeway divergence datum and search planning under spatial constraints that included land, illustrating the importance of operational constraints [52].

3.6. Evolution of Search Planning Objectives

Search planning objectives have expanded from POC/POD/POS to cost, coverage, path length, reward, energy, and workload balance. This diversification reflects the emergence of distinct planning tasks, including search-area allocation, route generation, sequential decision-making, and heterogeneous resource coordination. Appropriate task-specific metrics remain necessary. Their interpretation requires a clear statement of the decision being evaluated and a separate assessment of how changes in upstream probability information affect the planning metrics and operational outcomes. In a Canadian search-planning context, Laperrière-Robillard et al. used supervised learning to approximate simulation-based POS, demonstrating rapid AI-based estimation of classical search effectiveness [91]. Many coverage and RL studies examined in this review instead used reward or coverage ratio as their final metric, consistent with their specific objectives.

For an ML-based POS surrogate, low average prediction error is insufficient to establish better constrained planning. Errors near competing high-quality feasible plans can reverse their ranking, and an optimizer may preferentially select plans whose POS is overestimated. Evaluation should therefore retain the same endurance, arrival-time, sensor, and exclusion-area constraints, check feasibility separately, and re-evaluate selected plans using the reference POS model or a separately specified simulator. Useful outcomes include plan-ranking agreement, reference-evaluated POS, constraint violations, and achieved plan quality within a common computation budget. Constraint-aware sampling in ship-design surrogate optimization provides an adjacent example of why the feasible region matters for surrogate construction [92]; it does not itself validate a POS surrogate. In the present framework, surrogate error and latency are component metrics, while improvement in the feasible plan selected using that surrogate is assessed as Decision Gain.

Table 2 summarizes the search-planning methods and their representative objectives discussed above.

Table 2. Search planning methods and their representative objectives.

Method Family	Main Decisions	Typical Objectives	Characteristics
Classical search theory/IAMSAR	Effort, pattern, spacing	POC, POD, POS, coverage factor	Direct operational interpretation
SAROPS/computer-assisted planning	Probability areas, effort allocation	POS, posterior probability	Integrates drift uncertainty and prior searches
Mathematical optimization/metaheuristics	SRU, area, and parameter allocation	Fitness, cost, coverage, POS	Handles complex constraints and multiple objectives
Coverage/path planning	Visit order, waypoints, routes	Coverage, path length, completion time	Evaluates movement efficiency; sensitivity to input probabilities requires separate validation
RL/DRL/MARL	Sequential policies	Reward, success rate, coverage; POS in some studies	Can adapt to dynamic environments; sensitive to reward design
Heterogeneous multi-SRU planning	Roles, routes, workload allocation	Energy, workload, risk, detection probability	Incorporates operational constraints; end-to-end comparison requires an integrated benchmark

4. Coupling Drift Prediction with Search Planning

Sections 2 and 3 show that drift prediction and search planning have developed distinct technical traditions and evaluation metrics. In MSAR, probability representation connects these streams. A planner uses predicted trajectories converted into a particle distribution, PDF, POC map, or search area; RMSE itself is not a planning input.

4.1. From Predicted Trajectories to Search Areas

In deterministic prediction, search areas can be expanded geometrically using datum and divergence, as formalized in IAMSAR operational guidance [9]. In probabilistic models, Monte Carlo particles or ensemble members form a spatial distribution that can be converted into search areas using particle density or confidence regions [12,16]. HDRs provide a more general distributional summary by including higher-density regions first [87]. A search area is an operational object intended to contain a specified amount of target probability, rather than simply an area with low prediction error.

The OpenDrift study in the southern coastal waters of Korea illustrates this distinction. Its implementation of IAMSAR-based area construction yielded high containment and a large search area, whereas the OpenDrift configuration produced smaller areas and lower containment. This is a containment–area trade-off within that implementation and does not establish the superiority of all IAMSAR or OpenDrift configurations. The study used LPC and CER alongside RMSE and NCLS to evaluate this trade-off [83], providing a bridge between trajectory accuracy and search effort.

4.2. Probability Maps as the Prediction–Planning Interface

Let p_i denote the target probability in cell i and d_i the conditional detection probability associated with the planned effort. The expression $\sum_i p_i d_i$ in Equation (2) provides a simple link between prediction and planning. The prediction model primarily changes the location and spread of target probability, whereas the planner determines the spatial distribution of detection probability under resource constraints. Thus, changing p_i can change the optimal plan even with the same planner, while improving the planner can change d_i for a fixed p_i.

Studies that incorporate probability maps into their objectives explicitly use this interface. Ren et al. used particle proportions as POC [36]; POS-DQN incorporated success probability into reward design [45]; and probability-aware UAV/USV planners generated routes that prioritized high-probability regions [44,48]. Optimizing only path length or uniform coverage may fail to transmit the forecast's probability structure fully to the planner.

4.3. Sequential Prediction–Planning Frameworks

Many integrated studies use the sequential pipeline in Equation (3):

$$\text{Environment} \rightarrow \text{Drift Prediction} \rightarrow \text{Probability Map} \rightarrow \text{Search Plan} \rightarrow \text{Evaluation}. \quad (3)$$

This structure permits upstream errors to affect downstream decisions, but its implementation alone does not identify which component causes a reported gain. Section 4.6 distinguishes implemented links from intervention evidence in the selected cases.

GA-based search-area deployment directly used particle distributions in fitness evaluation [51]. Jeong et al. extended this type of coupling by transforming time-resolved Monte Carlo particle distributions into weighted heatmaps for GA-based SRU deployment and then evaluating dynamically updated search plans using cumulative POS [90]. Probability-guided coverage planning also sequentially coupled drift prediction with path generation [44], while Laperrière-Robillard et al. approximated simulation-based POS with an AI surrogate to reduce latency in planning support [91]. These cases implement the prediction–probability–planning interface in different ways. However, for the cases in Table 3, evidence of a partial intervention, such as a probability update, must be distinguished from evidence isolating the downstream effects of alternative prediction models under a common planner and evaluation criteria.

4.4. Dynamic Updates and Closed-Loop Planning

An operational search rarely ends with a single plan. An unsuccessful search, a new wind or current forecast, an additional observation, or sensor contact requires the posterior probability to be updated. If d_{ik} denotes the detection probability for search k in area i, cumulative detection under an independence approximation is given by Equation (4)

$$D_i = 1 - \prod_k (1 - d_{ik}). \quad (4)$$

With fixed p_i, cumulative POS can then be written as $\sum_i p_i D_i$. Equation (4) assumes independence among detection events in repeated searches. In a moving-target search, the posterior $p_i(t)$ itself changes after unsuccessful searches and forcing updates; this approximation must therefore be distinguished from a full sequential Bayesian treatment of POS.

Classical search theory and SAROPS treat unsuccessful searches as information and update the posterior accordingly [7,8,10,11]. Ren et al.'s Bayesian updating and dynamic POC path planning implement the same principle [36]. By contrast, many AI planning studies reviewed here held the initial probability map fixed within an episode or separated target-distribution updates from policy updates. Future closed-loop systems require AI prediction and planning to interact continuously on a shared timeline.

4.5. Artificial-Intelligence-Integrated and Emerging End-to-End Approaches

AI can support both sides of the interface. AI-based prediction can estimate p_i more accurately or quickly, while AI-based planning can allocate search effort to improve the resulting detection probabilities d_i for a given p_i. Connecting an uncertainty-aware neural forecast, a surrogate POS estimator, and an RL planner in one loop could further

reduce observation-to-decision latency. Probabilistic DNNs and neural ocean-transport models provide foundations for uncertainty-aware drift prediction [31,32]. In adjacent weather and environmental forecasting, ML surrogates integrated with data assimilation and observation-to-forecast architectures are developing rapidly [80–82], offering technical directions relevant to future MSAR prediction–planning systems. On the planning side, DRL, multi-agent RL (MARL), and RL-assisted resource allocation provide sequential decision capabilities [42,45,49].

The term "end-to-end AI" nevertheless requires care. Within this review's scope, evidence of prediction networks and planners jointly trained under a common objective and validated through final POS is limited. "AI-enabled integrated framework" or "closed-loop AI-assisted SAR" more accurately describes the current state of these systems.

4.6. Comparison of Existing Integrated Studies

Table 3 compares the purposively selected cases described in the Review Scope and Literature Selection subsection. P denotes Prediction; I, Information/Probability representation; D, Decision/Planning; and O, Operational outcome. Implemented stage connections are distinguished from validated effects. The presence of P does not establish an improvement in prediction accuracy, and model-based or simulated O is not equivalent to independent field validation. The table does not rank algorithms or exhaustively enumerate eligible studies.

Table 3. Illustrative cross-stage evidence map of purposively selected integrated studies. Stage labels indicate implementation scope; the final column describes identified intervention evidence and its limits. Model-based or simulated O does not imply independent field validation. Section, table, and figure locations in the cells refer to the cited studies.

Study	Stages Linked	Coupling and Evaluation Basis	Evidence Assessment
Breivik and Allen [12]	P–I	Ensemble PDFs/search areas; trajectory and area comparison	Implemented coupling; P–I evaluation only.
SAROPS [10,11]	P–I–D–O	Operational system; model-based POS	Implemented coupling; operational use does not isolate predictor effects.
Ren et al. [36]	P–I–D–O	POC updates and routes; simulated success/search time	Partial intervention evidence: probability updating (Section 5.3.2 of Ref. [36]); not alternative predictors.
Hong and Kim [51]	I–D–O	GA deployment; model-based fitness/detection evaluation	Implemented coupling; no common-planner predictor comparison.
Jeong et al. [90]	P–I–D–O	Dynamic particles/heatmaps; simulated cumulative POS	Partial intervention evidence: planning comparison and ablations; predictor effects not isolated.
Wu et al. [44]	P–I–D	Hierarchical maps and DRL; error/coverage/path metrics	Implemented coupling; complete predictor-intervention comparison unconfirmed.
Liu et al. [45]	I–D–O	POS-DQN; numerical POS/reward/search efficiency	Partial intervention evidence: planner comparison; predictor effects not isolated.
Zhang et al. [48]	P–I–D	Drift probabilities and aerial coverage; path metrics	Implemented coupling; complete predictor-intervention comparison unconfirmed.
Saito [52]	I–D–O	Datum/land constraints; model-based POS	Partial intervention evidence: practical constraints; predictor effects not isolated.
Kim et al. [83]	P–I	Area construction; observed containment/trajectory metrics (Section 4.2; Table 10; Figure 15 in Ref. [83])	Partial intervention evidence: P–I comparison; no executed search outcomes.

The evidence is interpreted using the rules in Table 4. A complete prediction-intervention comparison must satisfy all of the listed comparison conditions. Jointly trained prediction and planning constitute a distinct integrated approach; their total effect does not by itself isolate the prediction-side contribution.

Table 4. Interpretation rules for the evidence map in Table 3. These rules concern attribution of effects, not overall study quality.

Assessment	Interpretation rule
Implemented coupling	An output from one stage is used by another. This establishes a connection, not the effect of improving that output.
Partial intervention evidence	A comparison addresses selected stages, such as a probability update or planner change; its conclusions are restricted to the intervention evaluated.
Complete prediction-intervention comparison	Alternative predictors or prespecified prediction perturbations are compared using a common planner, resource/sensor constraints, and evaluation conditions, with probability representation, decision quality, and mission outcomes reported.
Unconfirmed	The examined evidence does not establish whether the complete comparison was performed; this denotes uncertainty in assessment, not proof of absence.

The cases illustrate different units of validation. Kim et al.'s containment–area comparison addresses the P–I interface but does not evaluate planner execution [83]. Ren et al.'s probability-update comparison extends the evaluation to simulated search outcomes and provides partial I–D–O evidence, but it does not isolate gains from alternative predictors [36]. Jeong et al.'s planning-side comparison and component ablations likewise provide partial I–D–O evidence while leaving predictor-side effects unisolated [90]. Liu et al.'s POS-DQN comparison evaluates planning-side changes using probability-based objectives [45]. Together, these cases show why evidence from one intervention cannot automatically be attributed to another stage. The narrower gap concerns comparisons that trace alternative-predictor effects under common planning and evaluation conditions. This observation is limited to the selected evidence; unconfirmed assessments do not establish research absence. Remaining uncertainty in those assessments also limits the strength of the synthesis.

5. Synthesis: How Do Prediction Improvements Translate into Search Planning and Operational Outcomes?

The distinction between implemented coupling and attributed effects in Section 4.6 motivates three research questions (RQs):

- RQ1: How have the evaluation objectives of drift prediction and search planning diverged, and what information does each metric preserve or lose for downstream SAR decisions?
- RQ2: Through which probability representations are drift prediction outputs transmitted to search-area generation, SRU allocation, and path planning?

- RQ3: To what extent have existing studies evaluated how improvements in drift prediction, particularly AI-based improvements, affect operational outcomes through probability representation and planning decisions?

These questions provide a framework for tracing whether an improvement at one stage produces a meaningful change at the next while retaining evaluation methods appropriate to each stage.

5.1. Stage-Specific Metrics Preserve and Lose Different Information

Drift prediction has expanded from deterministic trajectories to ensemble and probabilistic forecasts, ML/DL, hybrid physics–AI approaches, and uncertainty-aware prediction. Evaluation has broadened from distance error to NCLS and skill scores, distributional calibration, containment, and effort. Search planning has expanded from classical POC–POD–POS formulations to optimization, coverage planning, RL/DRL, and multi-agent coordination, with objectives that include coverage, reward, time, energy, and workload balance.

The answer to RQ1 is that differences in metrics reflect differences in evaluation objectives. Drift prediction estimates target states and uncertainty, whereas search planning seeks effective decisions given probability information and resource constraints. Pointwise RMSE captures the distance error of a representative position but does not directly measure probability ranking or calibration. Coverage and path length describe movement efficiency but may not represent the combination of target probability and sensor detectability. POS incorporates both probability and detection, but it is difficult to use alone to identify which upstream component produced an improvement. Stage-specific metrics should therefore be retained while explicitly reporting which information is transmitted to the next stage and which information is lost when moving between metrics.

5.2. Probability Representation Mediates Prediction and Planning

Prediction accuracy is not transmitted directly to a planner. The interface is a probability representation, such as particles, a PDF, a POC map, a search area, or a priority field. An AI drift model may reduce RMSE without changing the plan if the ranking, spread, or calibration of p_i changes little. Conversely, a small change in mean error may lead the same planner to select a substantially different route if the locations and rankings of high-probability regions change.

Evaluation therefore needs to examine both the proximity of the mean track to observations and the position of the actual target within the probability field. This is consistent with the emphasis on evaluating calibration and sharpness separately in probabilistic forecasting [85]. Rank histograms and multivariate ranks assess the position of observations relative to an ensemble distribution [84,86], while HDRs prioritize higher-density regions [87]. These concepts inform the evaluation of operational priorities within MSAR probability maps.

5.3. The Evidence Gap Concerns Cross-Stage Impact Validation

The selected integrated studies include P–I–D–O implementations and partial validation of the downstream effects of probability updates. The answer to RQ3 is therefore that evidence isolating the effects of alternative predictors under a common planner and evaluation criteria, with intermediate changes reported, remains limited within the selected cases. Partial intervention evidence is acknowledged while distinguishing the attribution of effects across the full pipeline. This observation is not extended to a field-wide claim of absence or an estimate of prevalence.

Forecast science distinguishes forecast consistency, quality, and value: improved forecast quality need not increase its value for a particular decision problem [93]. Cost–loss analysis and ensemble-forecast evaluation further show that value depends on the user's decision structure and is distinct from forecast skill [94,95]. Predict-then-optimize describes a prediction-to-optimization pipeline, whereas decision-focused and task-based learning use downstream objectives to guide training [96–98].

The four-stage Gain framework in Equation (5) organizes evaluation and attribution across an MSAR pipeline. It applies to jointly trained systems and separately developed predictors and planners, making the probability/POD interface, resource constraints, and distinction between plan quality and time-dependent execution outcomes explicit. It is an evaluation scheme rather than a new learning algorithm; the arrows identify interfaces to evaluate and do not imply positive or monotonically transmitted gains.

$$\text{Prediction Gain} \rightarrow \text{Information Gain} \rightarrow \text{Decision Gain} \rightarrow \text{Operational Gain}. \tag{5}$$

Prediction Gain is an improvement in the prediction model itself, measured by quantities such as RMSE, an NCLS-based skill score, or inference time. Prediction generates a future target state or distribution. Information/Probability representation transforms or summarizes that output into particles, grid probabilities, search areas, or other inputs consumed by the planner. For a probabilistic predictor, this transformation can be an identity mapping; the two stages need not be separate modules. Information Gain is an analytical label for improvements in probability ranking, containment, or search-area efficiency while maintaining or improving calibration. It does not denote an information-theoretic quantity such as entropy reduction or Kullback–Leibler divergence.

One quantitative operationalization of Information Gain uses the reduction in a prespecified proper probability score on common held-out cases. For N evaluation cases and M mutually exclusive spatial categories, let p_{mni} be model m's probability for cell i in case n, and let y_{ni} equal one for the cell containing the observed target and zero otherwise. Equation (6) defines a multicategory Brier score and its improvement:

$$\begin{aligned} B_m &= \frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{M}(p_{mni}-y_{ni})^2, \\ G_I^{(B)} &= B_{\text{base}} - B_{\text{new}}. \end{aligned} \tag{6}$$

Positive $G_I^{(B)}$ indicates improved probability-score performance; zero and negative values indicate no change and deterioration, respectively. Cases must use the same forecast horizon and spatial partition, with an explicit outside-domain category when required so that the scored probabilities sum to one. This is one operationalization, not a new proper score or a universal scalar measure of search usefulness [85]. Calibration, containment, search-area efficiency, and priority diagnostics should also be reported where relevant. A lower Brier score alone neither establishes calibration improvement in every respect nor implies Decision Gain.

Decision Gain denotes improved plan quality under common constraints and the same prespecified evaluation criterion, rather than a change in allocation, search order, or path alone. Operational Gain denotes improved mission outcomes when the selected plan is executed or assessed in a separately specified simulation. POS, detection time, and resource-hours may be used at both stages, but the same internally computed POS must not be counted as independent evidence of both gains. Model-based expected performance, simulation outcomes, and field outcomes represent distinct levels of evidence.

Operational Gain should always be qualified by its evidence source: simulated Operational Gain compares execution outcomes in a specified simulator; field-exercise Oper-

ational Gain uses controlled trials; and mission-based Operational Gain uses actual SAR operations. These levels should not be pooled or treated as interchangeable. Use of real drifter observations validates aspects of prediction or containment but does not by itself establish mission-based Operational Gain. Likewise, deployment of a planning system establishes operational use, not a causal benefit from changing its predictor.

For a maximization problem, let $J_{\text{ref}}(a)$ be a common evaluation function, $\mathcal{A}$ a common feasible set, and a_m the plan produced by a planner using predictions from model m. Defining $J^{\star}_{\text{ref}} = \sup_{a\in\mathcal{A}} J_{\text{ref}}(a)$ and $R_{\text{ref}}(a) = J^{\star}_{\text{ref}} - J_{\text{ref}}(a)$, Decision Gain can be expressed as Equation (7)

$$\begin{aligned} G_D &= R_{\text{ref}}(a_{\text{base}}) - R_{\text{ref}}(a_{\text{new}}) \\ &= J_{\text{ref}}(a_{\text{new}}) - J_{\text{ref}}(a_{\text{base}}). \end{aligned} \tag{7}$$

Comparisons must use a common evaluation distribution, detection model, time horizon, and definitions of resources and costs. When the true distribution is unknown, the assessment must be identified as evaluation against held-out cases or an independently specified simulation or evaluation model. If the optimum is unknown, absolute regret need not be reported: the difference in plan quality remains available because the common optimal benchmark cancels.

For a numerical illustration of Equation (7), consider two search regions with a common evaluation distribution $q = (0.6, 0.4)$. Suppose two feasible plans use the same resource budget and have conditional detection probabilities $d_{\text{base}} = (0.4, 0.8)$ and $d_{\text{new}} = (0.8, 0.4)$; assume equal regional effort costs and a detection–effort model under which both allocations are feasible. Using $J_{\text{ref}}(a) = \sum_i q_i d_i(a)$ gives $J_{\text{ref}}(a_{\text{base}}) = 0.56$ and $J_{\text{ref}}(a_{\text{new}}) = 0.64$, so Decision Gain is 0.08, or eight percentage points of reference-evaluated detection probability. This arithmetic example explains plan-quality evaluation only: it does not establish a prediction-accuracy improvement, report an empirical MSAR result, or provide independent evidence of Operational Gain.

Improvements do not automatically propagate through these stages. For illustration, a hypothetical 20% reduction in deterministic pointwise RMSE may produce the same search region, and a fixed-pattern planner may generate an unchanged route. Conversely, a small shift in probability mass toward a high-priority region may cross a discrete allocation boundary and change SRU assignments. A changed decision is not necessarily a better decision. Nonlinear thresholds and interactions may therefore arise between component-level and operational gains. Sensitivity analysis should identify the level of prediction or information improvement that changes decision quality and the user requirements, resource constraints, and sensor conditions under which that change creates operational value.

The evidence gap identified here concerns attribution, not proof that more accurate prediction has no operational benefit. Within the selected cases, the available comparisons do not sufficiently isolate alternative-predictor effects through all stages under common conditions. The hypothetical example above illustrates a possible mechanism; it is not a measured counterexample or an estimate of how often prediction improvements fail to benefit MSAR.

5.4. The Prediction-to-Planning Cross-Stage Validation Gap

Drift prediction, probability representation, coverage planning, and RL appropriately use different metrics because their roles differ. What remains necessary is linked reporting of how a performance change at one stage alters the information and decisions at the next. Evaluation should sequentially establish whether lower RMSE produces a more useful probability map, whether that map changes allocation or routing under the same planner, and whether those changes improve POS or detection time. Classical POS is useful for

interpreting system-level outcomes because it combines probability and detection, but it must be considered alongside prediction calibration, sensor models, overlap, cumulative detection, and timing. Operational value also includes arrival time, expected detection time, endurance, resource-hours, mission cost, risk, communication constraints, and the human coordinator's assessment of feasibility.

Cross-stage evaluation therefore requires metrics for four corresponding levels: trajectory accuracy, such as RMSE and NCLS; probability quality, such as calibration, containment, HDRs, and probability ranks; decision quality, such as plan-quality or regret differences under a common criterion; and operational effectiveness, such as POS, detection time, and resource-hours measured during execution or in a specified simulation. These metrics measure different changes along the pipeline. Kim et al.'s use of LPC and CER alongside trajectory metrics is a step toward such multilevel evaluation [83].

5.5. Separating Prediction-Side and Planning-Side Effects

An experimental design with at least four combinations is useful for separating AI-related effects. Table 5 presents the corresponding factorial comparison.

Table 5. A 2×2 comparison separating the effects of prediction AI and planning AI.

	Baseline Planner	**Advanced/AI-Assisted Planner**
Baseline prediction	A: Baseline system	B: Evaluation of planning-side gains
Advanced/AI-assisted prediction	C: Evaluation of prediction-side downstream effects	D: Evaluation of integrated/end-to-end gains

Comparing A and C reveals changes in information, decision, and operational metrics when only the prediction-side intervention changes. Comparing A and B isolates the planning-side intervention given the same probability information. The difference between D and A represents the total effect of the advanced or AI-enabled system, but B and C are needed to distinguish prediction-side from planning-side contributions. For a common scalar outcome Y, a factorial interaction can be defined as $\Delta_{\text{int}} = (Y_D - Y_C) - (Y_B - Y_A)$ to assess whether the combined effect differs from the sum of the separate effects. Such 2×2 ablation and interaction analyses are particularly useful for systems combining ML drift models with DRL or optimization-based planners. All four combinations must use the same outcome definition, evaluation cases, horizon, resource constraints, and computation-budget rules. Comparing only each system's internally predicted POS risks mistaking changes in forecast confidence for actual gains.

5.6. Probability of Success as a Common Operational Reference

Classical POS provides a common operational language for prediction and planning. Predictor outputs change p_i, while the planner changes d_i. The structure $\sum_i p_i d_i$ can therefore serve as a common reference for interpreting system-level effects regardless of whether ML, GAs, or DRL are used. Its role here is to connect component metrics to operational interpretation, not to introduce a new mathematical formulation. Modern AI studies should report the relationships between component metrics and POS, detection time, and resource metrics, rather than relying on POS alone.

Model-computed POS is conditional on the predicted target distribution and the assumed detection model. Comparisons must distinguish internally predicted POS from performance on common held-out cases or under a common evaluation distribution and detection model. For example, changing a two-cell forecast from $(0.9, 0.1)$ to $(0.99, 0.01)$ does not change the evaluated success probability if both plans search only the first cell in the same way and the common evaluation distribution is $(0.5, 0.5)$. Dynamic missions should

report both detection probability by a common horizon and detection time, specifying the proportion of undetected cases and how they are handled in time statistics. Detection probability must not be equated with successful rescue or survival probability.

6. Future Research Directions

Three immediate priorities follow: controlled comparisons on common cases, probability evaluation linked to decisions, and realistic execution validation. Table 6 specifies a minimum reporting protocol; the existing 2×2 design separates prediction-side and planning-side effects.

Table 6. Minimum protocol for cross-stage evaluation.

Step	Required Specification or Output
Common cases	Same held-out targets, initial uncertainty, forcing availability, horizon, spatial partition, sensor model, resources, and computation-budget rules.
Intervention	Identify the changed predictor, input/fusion method, probability transformation, or planner; retain the other components in the primary comparison.
Prediction Gain	Report a signed change in the prespecified prediction metric, with its direction of improvement.
Information Gain	Report a proper-score change and relevant calibration, containment, area, or priority diagnostics on the same cases.
Decision Gain	Re-evaluate both feasible plans under a common reference criterion; report the paired difference and constraint violations.
Operational Gain	Evaluate execution separately; identify simulation, field exercise, or actual mission, and report detection/time/resource outcomes and uncertainty.

6.1. Priority 1: Common Benchmarks and Controlled Comparisons

Benchmarks should compare the four combinations of conventional/AI prediction and conventional/AI planning using common drift scenarios and SRU configurations. A public benchmark should provide environmental forcing, initial uncertainty, target types, observation tracks, sensor/POD models, SRU constraints, and computation budgets. Reporting should jointly cover prediction RMSE/NCLS, probability calibration and containment, planning coverage and paths, cumulative POS, expected detection time, and computation latency.

Attributing prediction-side effects requires explicit specification of the spatial and temporal uncertainty of the initial datum, scenario-mixture weights, target types, and sensor assumptions, including sweep width. The initial datum should not necessarily be treated as a fixed point with zero uncertainty, and sweep width is not simply a geometric sensor footprint. These conditions should be fixed in the primary comparison, with their effects examined in separately identified sensitivity analyses. Because they can interact with prediction improvements and change the preferred plan, reported gains are conditional on the initial scenarios, detection capabilities, and time horizon and should not be generalized automatically to other settings.

One feasible implementation could use OpenDrift as an open trajectory-generation and ensemble layer [16], with IAMSAR-based standard patterns and effort allocation as a conventional planning baseline [9]. Publicly shareable grid-based optimization, GA, or RL planners could then use the same POC/POD interface, creating a reproducible testbed in which prediction and planning can be replaced independently. SAROPS is appropriately used as an operational reference for the integration of posterior updates and

search-effort allocation in decision support, rather than as a public implementation of such a benchmark [10,11].

The operational sensitivity to prediction improvements should be quantified. Artificially reducing prediction RMSE by 5%, 10%, or 20%, or varying probability spread and calibration incrementally, could identify when search areas, allocation boundaries, and POS change meaningfully. This would help characterize the accuracy needed to change a search decision. Perturbations should be designed to avoid unrealistic distortion of calibration, and Decision Gain and Operational Gain should be reported separately.

6.2. Priority 2: Probability Quality and Search-Oriented Evaluation

RMSE, MAE, and NCLS describe discrepancies between predicted trajectories and observations but do not directly indicate the search priority assigned to the observed location within the predicted probability field. LPC aggregates time-specific binary indicators of whether the observed location lies within the predicted area, while CER relates containment to search area [83]. Neither distinguishes between a target in the highest-density cell and one in a low-density cell within the same predicted area.

Normalized Search Priority Rank (NSPR) is considered here as an auxiliary diagnostic whose practical value requires validation. At time t, define a searchable domain of $M > 1$ equal-area cells, excluding land and other masked or inaccessible cells; regrid to equal-area cells when necessary. Let $p_{i,t}$ be the probability conditional on this domain, with $\sum_{i=1}^{M} p_{i,t} = 1$. Report the retained probability mass before normalization and observed out-of-domain rate separately. An out-of-domain observation is a containment failure, not an assigned worst rank; zero retained mass makes the conditional diagnostics undefined. For an in-domain target in cell i_t^*, Equation (8) gives its rank in decreasing probability order:

$$r_t = 1 + \sum_{j=1}^{M} \mathbb{I}(p_{j,t} > p_{i_t^*,t}) + \frac{\sum_{j=1}^{M} \mathbb{I}(p_{j,t} = p_{i_t^*,t}) - 1}{2}. \tag{8}$$

Equation (8) uses midranks for ties; Equation (9) normalizes the rank:

$$\mathrm{NSPR}_t = \frac{r_t - 1}{M - 1}, \tag{9}$$

Values near zero indicate high target priority and values near one low priority. NSPR is not the fraction of area searched: under an idealized ordering of equal-area, equal-cost cells with random tie-breaking, the expected fraction visited before the target is $(r_t - 1)/M$. Neither quantity directly measures detection time with imperfect detection, travel costs, obstacles, or vehicle dynamics.

NSPR complements established forecast-verification tools, including ensemble rank histograms [84] and multivariate ranks and energy scores [86]. It adds a search-priority interpretation but is not a proper scoring rule; probability quality still requires calibration and sharpness assessment [85].

Rank alone discards differences in probability magnitude. It is therefore useful to report the cumulative probability mass of searchable cells whose probabilities are at least as high as that of the observed target cell, as defined in Equation (10):

$$H_t = \sum_{j:p_{j,t} \geq p_{i_t^*,t}} p_{j,t}. \tag{10}$$

On an equal-area grid, H_t is the conditional mass of the density superlevel set containing all cells with probability at least as high as the target cell, including ties. It has a discrete HDR interpretation [87] and, for a Bayesian posterior, a highest posterior density

(HPD) interpretation. For example, $H_t = 0.2$ denotes 20% of the in-domain probability mass; ties may prevent exact probability levels. NSPR measures spatial rank, whereas H_t measures mass above a density threshold. A uniform distribution gives $H_t = 1$, so it is not a universally lower-is-better score.

Table 7 illustrates these diagnostics using five equal-area cells. All three targets are inside the domain, so binary containment equals one despite different priorities. The example is synthetic, not empirical validation.

Table 7. Five-cell synthetic illustration of NSPR and H_t.

Case	Cell Probabilities	r_t	NSPR	H_t
A	[**0.35**, 0.25, 0.20, 0.10, 0.10]	1	0.000	0.35
B	[0.30, **0.25**, 0.20, 0.15, 0.10]	2	0.250	0.55
C	[0.40, 0.25, 0.15, **0.10**, 0.10]	4.5	0.875	1.00

Note: Bold values indicate the probability assigned to the observed target cell. The target ranks first in A, second in B, and jointly last in C.

Table 8 summarizes the recommended multilevel evaluation metrics for the prediction-to-search pipeline.

Table 8. Recommended multilevel evaluation of the prediction-to-search pipeline, including auxiliary diagnostics.

Evaluation Level	Example Metrics	Question Addressed
Trajectory accuracy	RMSE, MAE, NCLS	How close is the representative trajectory to the observed position?
Distribution quality	Calibration, proper scores, energy score, HDR-based coverage/mass	Is the predicted distribution reliable and appropriately concentrated?
Operational containment	LPC, CER	Does the predicted area contain the actual target, and how much area is required?
Search priority (auxiliary)	NSPR, H_t	How highly is the actual target prioritized within the probability map?
Decision quality	Quality difference under a common criterion, regret reduction	Does plan quality improve under the same evaluation criterion?
End-to-end outcome	POS, detection time, resource-hours, mission cost	Does the probability information improve search outcomes and resource efficiency?

NSPR is sensitive to grid resolution and domain definition and omits physical distance, route connectivity, and travel time. Report it alongside H_t, trajectory errors, proper scores, containment/area diagnostics, and final search outcomes. Validation should test robustness across grids and relationships with search effort, detection time, and POS.

Low mean error can coexist with an overconfident distribution. Uncertainty-aware drift models should therefore be evaluated for calibration and sharpness under regional and seasonal shifts [85,86]. Probabilistic and physics-informed models [31,32,75] and assimilation with simplified models or ML surrogates [78,81] offer complementary directions.

6.3. Priority 3: Realistic Closed-Loop and Field Evaluation

A closed-loop system should incorporate unsuccessful searches, new drift observations, and updated forcing into the posterior before rerunning the planner. Combining classical Bayesian search and moving-target theory [7,8] with modern RL/MARL could support more realistic adaptive planning than a static probability map. Cumulative POD

and overlap must be tracked accurately, and real-time latency should be treated as a constraint as the target distribution evolves. Under the four-stage framework, each update cycle should assess whether Prediction Gain and Information Gain are reflected in the new posterior and subsequently translate into Decision Gain and Operational Gain.

Crewed vessels, aircraft, UAVs, USVs, and civilian vessels differ in speed, endurance, sensor footprint, and risk, making uniform coverage models insufficient for comparison. Existing studies have considered response time, coverage, allocation cost, workload, and SAR success in resource allocation and search planning [40,41,46,49,51]. Future work should explicitly incorporate SRU-specific sensor capabilities into POD and jointly optimize cumulative POS and routes. AI is also most realistically implemented as decision support that explains probabilities, uncertainty, and trade-offs among alternative plans to the Search and Rescue Mission Coordinator. SAROPS and ML-based POS surrogates illustrate the practical value of computational support centered on human operators [10,11,91]. Operational Gain should therefore encompass mission-level value, including whether an operator can understand, modify, and approve a plan within the available time.

7. Conclusions

This structured narrative review examines how drift prediction is connected to MSAR planning through probability representation. Its contribution is an MSAR-specific framework distinguishing Prediction Gain, Information Gain, Decision Gain, and Operational Gain, together with an illustrative evidence map and common comparison conditions. The framework applies established forecast-value and prediction-to-decision principles; it does not introduce a new learning algorithm or imply that gains necessarily propagate between stages.

The ten selected cases demonstrate implemented coupling and partial intervention evidence. The narrower unresolved issue is attribution: whether changes in a predictor improve the probability information supplied to a common planner, the quality of its feasible decisions, and separately evaluated execution outcomes. Limited evidence for this chain within the selected set must not be interpreted as a field-wide absence claim or as evidence that better prediction has no operational benefit. Model-based plan quality, simulated execution, field exercises, and actual SAR missions remain distinct evidence levels.

The immediate priorities are common-case controlled benchmarks, probability evaluation linked to plan quality, and realistic closed-loop execution validation. A practical assessment fixes the target cases, sensor/resource constraints, horizon, and computation-budget rules; changes specified components; and reports paired metrics at all four stages. Proper scores and containment measures should accompany trajectory error, while auxiliary diagnostics such as NSPR require further validation. This protocol makes the evidence needed to connect prediction improvements with detection outcomes and resource use explicit.

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